

SIGma Fall 2026
CG:SHOP Work Session

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Outline

CG:SHOP

Git/Env Setup



Section 1

CG:SHOP



What is CG:SHOP?

- **CG:SHOP** (Computational Geometry: Solving Hard Optimization Problems) is a computational geometry competition for students and researchers.
- Each year, teams build solvers for a difficult geometric optimization problem.
- The 2027 challenge is **Multi-Robot Lawn Mowing for Polyominoes**.

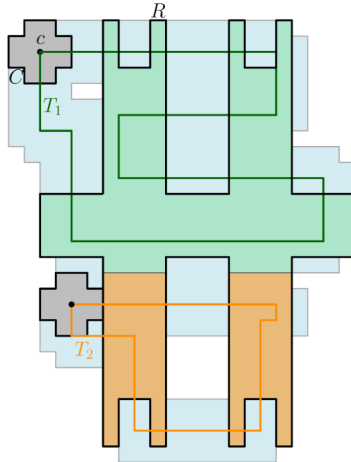


Problem Background

- A *polyomino* is a polygonal region made from grid cells whose boundary is horizontal and vertical.
- Each problem instance gives us a region R (the *field*), a cutter shape K , a center point $c \in K$, and a number of cutters C , where R and K are polyominoes.
- Each cutter moves its center along a closed, axis-parallel trajectory (will define later).
- Cutters do not block one another and may leave R .
- *Problem Statement:* Given a *field* R and C cutters with fixed shape, find a list of trajectories $T_1, T_2, \dots, T_C$ that cover the entire field, while minimizing the length of the longest trajectory.



An Example



An example polyomino region (light green & light orange), a small polyomino cutter with center (gray), a (suboptimal) pair of valid tours (dark green and orange) and the region it covers (light blue).



Formal Problem Statement

- **Input:** a polyomino region R , a cutter shape K with center c , and an integer C .
- **Solution:** exactly C closed trajectories $T_1, \dots, T_C$ for the cutter centers.
- A trajectory is a cyclic sequence of points $T_i = (p_{i,0}, \dots, p_{i,m_i-1})$ with $p_{i,j} \in \mathbb{Z}^2$ and axis-parallel edges.
- Its length is

$$\ell(T_i) = \sum_{j=0}^{m_i-1} \|p_{i,j+1} - p_{i,j}\|, \quad p_{i,m_i} = p_{i,0}.$$

- We require that every cell of R is swept by at least one translated cutter.
- **Objective:** minimize $\max_i \ell(T_i)$.
- Note: In case you were wondering, this is NP-hard.



CG:SHOP Logistics

- The competition starts on October 15, 2026.
- Final submissions are due January 31, 2027.
- Example instances and Python utilities are already available.
- Discord for CG:SHOP problems: discord.gg/3FXC6xzzWe.
- We might also have a professor advise us, this is still TBD.



CG:SHOP Logistics

- Please start taking a look at the [challenge website](#).
 - ▶ There are also many papers out there about similar problems.
- We will split into *algorithm* and *implementation* teams.
 - ▶ *Algorithm* teams will work on reading/understanding existing literature and generating/doing basic testing of potential algorithms.
 - ▶ *Implementation* teams will work on implementing the algorithms from the algorithm team as well as modifying existing algorithms.
 - ▶ Both teams need a strong understanding of current research in this area.
- Bring questions, observations, or solver ideas to the next meeting.



Project AI Policy

- You are free to use AI for understanding papers, setting up environments, or generating algorithmic ideas, but treat its output with caution.
- We will not be merging any code that we do not understand or is clearly AI-generated.
 - ▶ We need to understand the code that we write.



Section 2

Git/Env Setup

